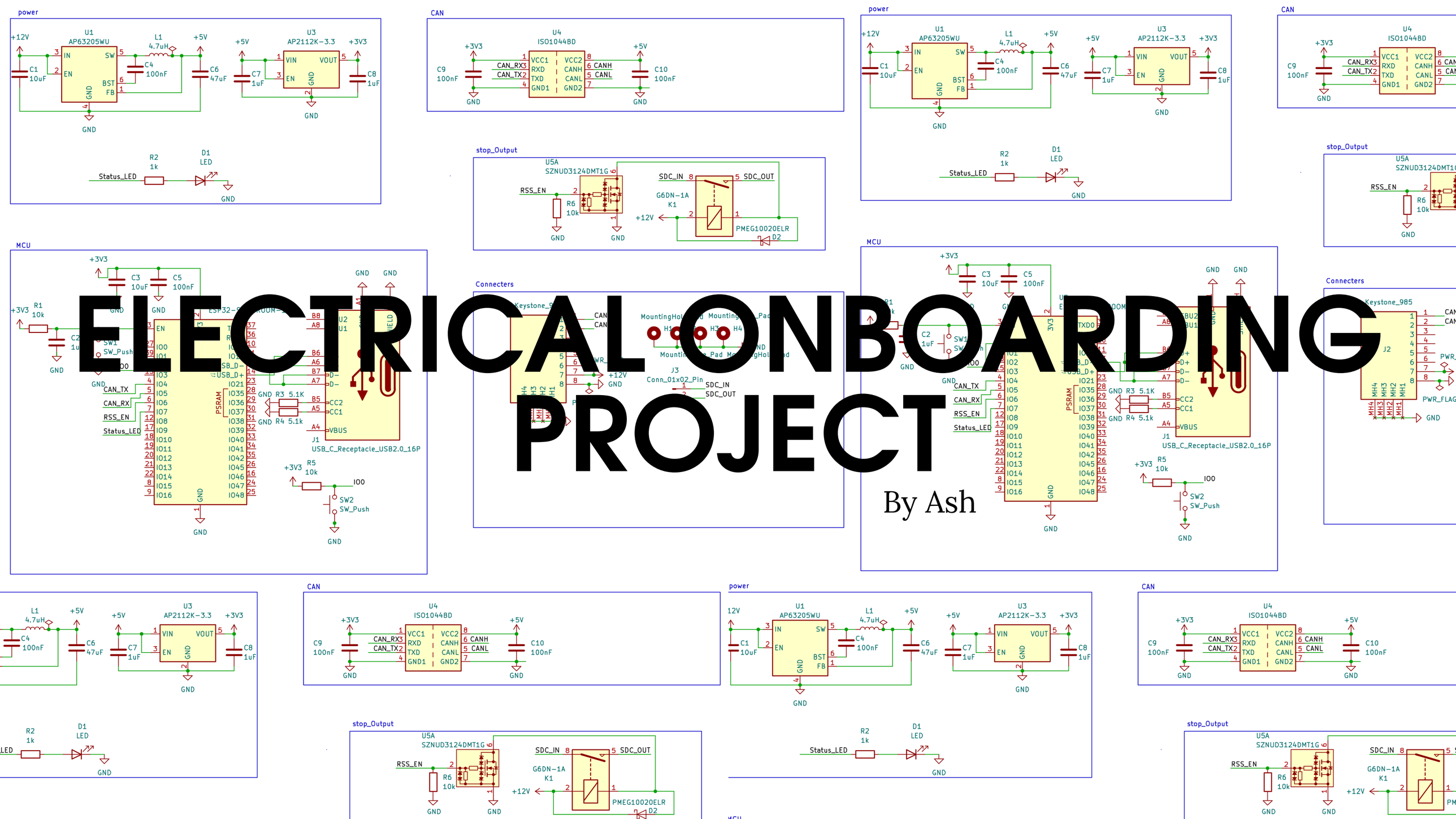
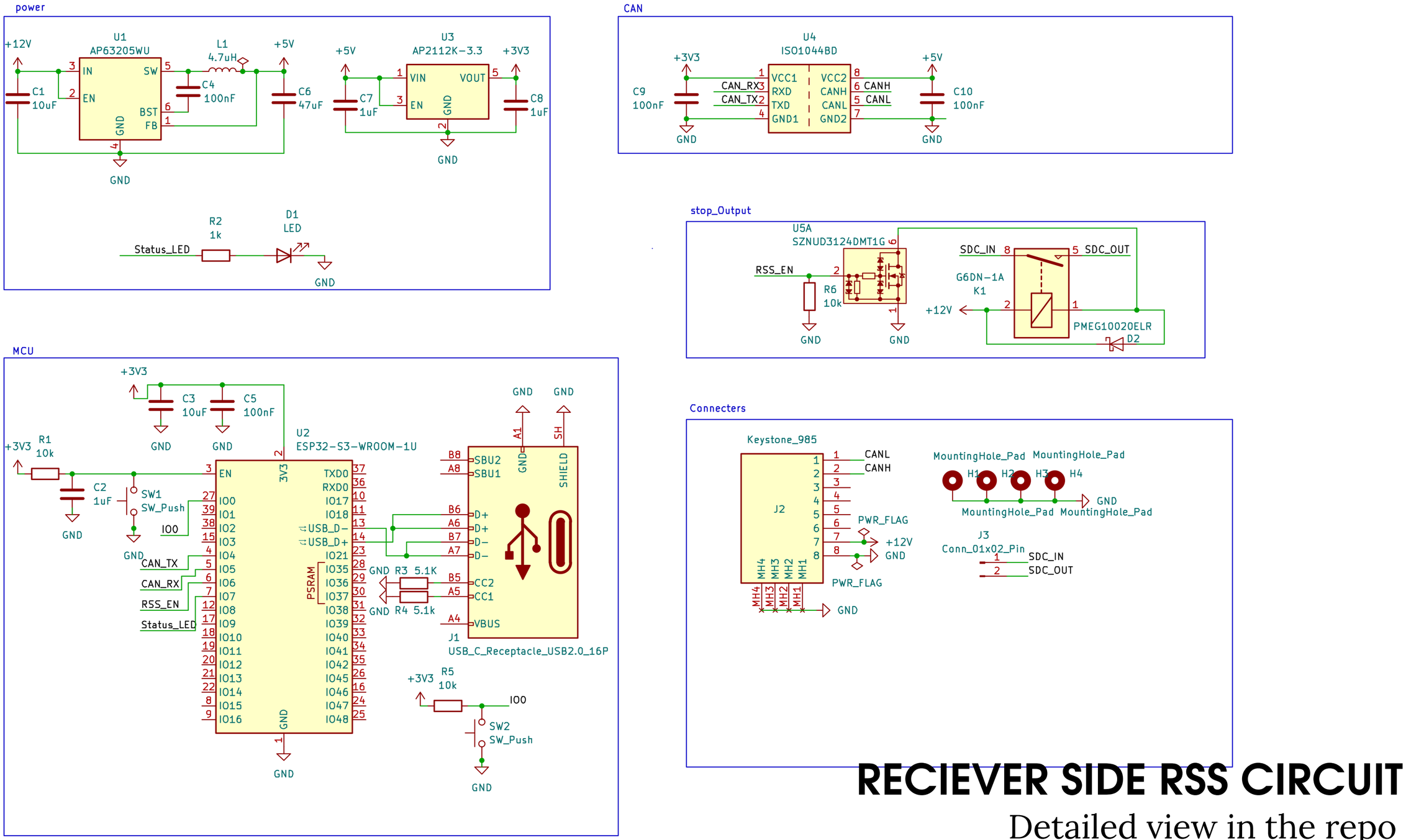

power
CAN
power
CAN
stop_Output
stop_Output
MCU
Connecters
MCU
Connecters
ELECTRICAL ONBOARDING
PROJECT
By Ash
CAN
power
CAN
stop_Output
stop_Output

WHAT I MADE:



RECIEVER SIDE RSS CIRCUIT

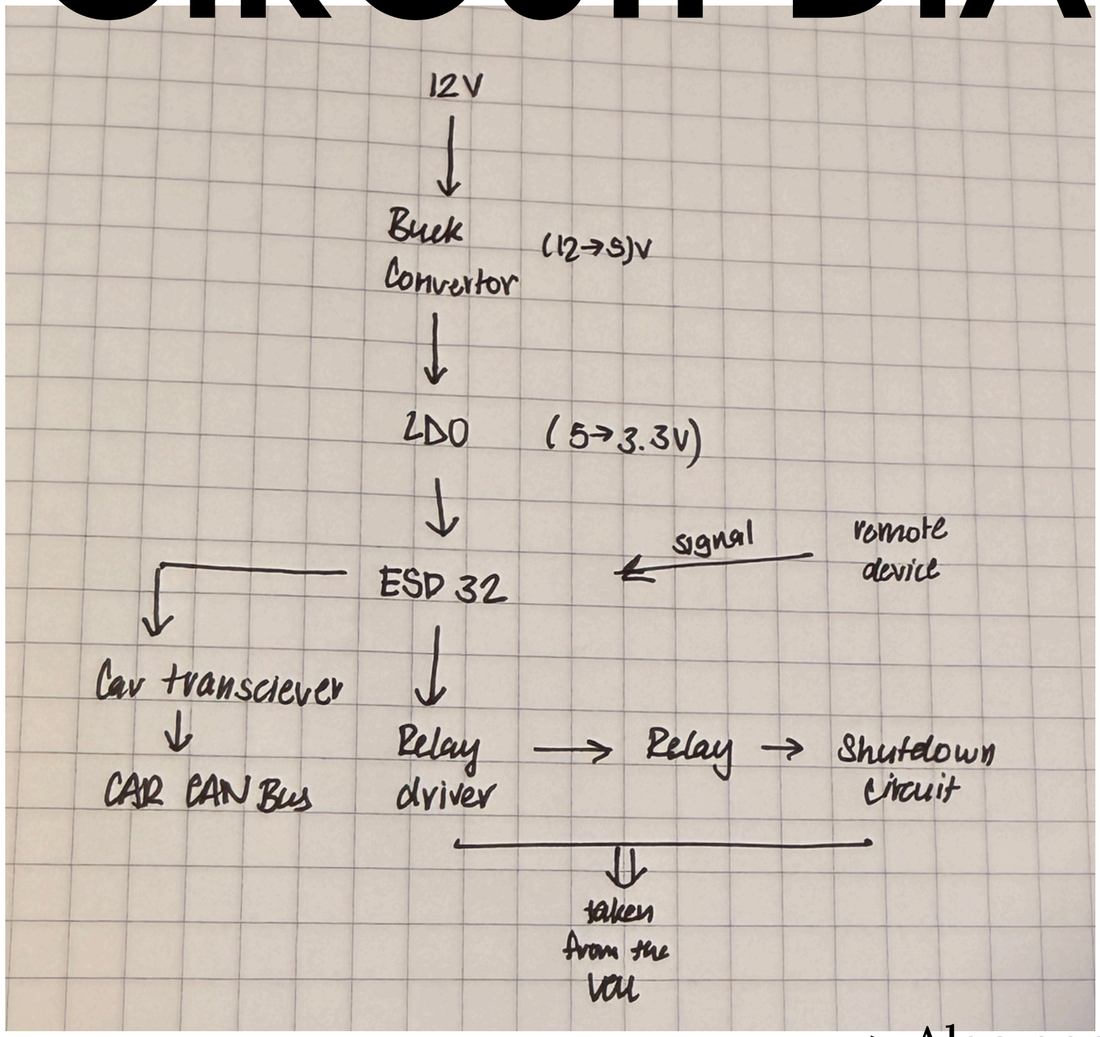
Detailed view in the repo

WHAT IT DOES?

- This is the receiver side of the radio for the RSS (Remote Stop System) that wold be placed on the car to open the shutdown circuit remotely

- This uses an ESP-32-S3-Wroom-U1 Module (as it supports external antennas, a requirement in the rules)

BASIC CIRCUIT DIAGRAM



> Also see my notes, they are cool.

COMPONENTS I FOUND INTERESTING

- ESP32-S3-Wroom-U1 Module (This was fun to use, way simpler than any other implementation I could find, and had lots of good examples online)
- I reused a lot of design blocks from the FS4 Library, like CAN Transceiver, Regulators, and even the shut down relay (I think the battery team uses the same one to shift from pre-charge to full mode?)
- Special mention to: Lowpass, I really like the concept cause I understood the concept and really love it

WHAT I LEARNT

- How to pick components
  - Figuring out specifications (Like the range of the radio, antenna specification)
  - Looking up implementations online
  - Figure out external modules requirement (how much voltage does this need, from where?) by reading the data sheet

- Lowpass, filter, gates, and pulldown (in a birds eye view sense)

- Reusing old circuit parts, like the blocks I mentioned in the previous slide, they are super helpful when designing something

THANK YOU!

Feel free to reach out for any questions!